

Presentation title:

Low-Cost Water Level Monitoring for Municipal Water Management

Presentation description:

This presentation showcases a low-cost, solar-powered water level sensor that enables continuous remote monitoring with sub-centimeter accuracy using long range radio or cellular communication. Designed to support municipal water management, the system delivers real-time data from field deployments across multiple waterways in the Augusta, GA region. We will highlight both the technical design and real-world performance of the device.

Program description:

Learn about a novel, low-cost water level sensor that uses buoyancy-based measurement and wireless data transmission to enable real-time, high-precision water monitoring for municipalities. Powered by an integrated solar panel and LiPo batteries, the device is designed for indefinite deployment in remote environments. Two versions are available: one transmits data via long-range radio, and the other uses the cellular Particle platform. This work is part of an ongoing research collaboration between Augusta University Department of Physics and Biophysics and the Augusta Utilities Department.

Bio:

Kathryn Dunstan is a senior at Augusta University, majoring in physics with a concentration in nuclear science. She works as a research intern in the Department of Physics and Biophysics, a position sponsored by the City of Augusta Utilities Department. Her research focuses on the development and testing of low-cost water quality monitoring devices, with the goal of supporting more effective and sustainable water resource management.

This presentation introduces a novel, low-cost water level sensor that utilizes buoyancy-based measurement and wireless data transmission for high-precision, real-time monitoring in municipal water systems. The device, powered by an integrated solar panel and LiPo batteries for indefinite deployment in remote environments, is available in two versions: one using long-range radio and the other utilizing the cellular Particle platform for data transmission. This research is part of an ongoing collaboration between Augusta University's Department of Physics and Biophysics and the Augusta Utilities Department.